

The selected-window race

a full-proof attempt and the surviving spectral-overlap gap

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Abstract

We attempt to prove that $\Delta_{q_{\text{arith}}}(k) < 1 - q_*(k)$ for infinitely many members of the named Lean-selected sequence $a_k = 2^k$, $r_k = \lfloor \sqrt{k} \rfloor$, $m_k = k2^{r_k} - 1$. The outcome is **PARTIAL**. We prove an exact spectral representation of the race and a genuine two-band sufficient criterion. The criterion gives a finite certificate at $k = 5$, but no uniform estimate: the raw resolvent norm bound grows from 4.82 to 4991.65 on $k = 5, \dots, 12$, and even the optimal two-band bound exceeds 1 from $k = 6$ onward. The fixed-mesh plateaux cannot supply an infinite subsequence because the r th plateau contains exactly $2r + 1$ indices. The precise unresolved statement is a source-derived estimate for the spectral measure of the explicit border vector near the saturating edge $\lambda = 1$. No Riemann-hypothesis claim, and no anti-RH claim, is made.

Binding outcome. PARTIAL: spectral reduction proved; norm route refuted; plateau route finite; spectral-overlap estimate open. The candidate race lemma is not proved, refuted, or promoted. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The exact object

Fix one selected finite window and its cap N . Use the positive part μ of the folded signed measure as the orthogonality measure. In its orthonormal polynomial basis let

$$B_{yi} = \sqrt{\nu_y} p_i(y), \quad C = B^T B, \quad u = \frac{b}{\sqrt{B_w}},$$

where ν is the negative part, b_i is the smooth-border functional on p_i , and $B_w > 0$ is the standing budget. These are exactly the matrices used by `deep_builder_probe.solve_qdag`. The finite augmented matrix in this frame is

$$\mathcal{A} = \begin{pmatrix} I - C & u \\ u^T & 1 \end{pmatrix}. \quad (1)$$

The r453/r457 identity gives

$$q_{\text{full}}^\dagger = q_* + \Delta_{q_{\text{arith}}}, \quad q_{\text{full}}^\dagger < 1 \iff \Delta_{q_{\text{arith}}} < 1 - q_*. \quad (2)$$

Equation (2) is scalar algebra; it does not by itself prove either inequality.

2 What can be proved

Theorem 1 (Spectral race identity). *Assume $C \prec I$. Let $Ce_j = \lambda_j e_j$ be an orthonormal eigendecomposition. Then*

$$q_{\text{full}}^\dagger = u^T (I - C)^{-1} u = \sum_{j=1}^N \frac{|\langle u, e_j \rangle|^2}{1 - \lambda_j}. \quad (3)$$

Moreover,

$$\mathcal{A} \succ 0 \iff q_{\text{full}}^\dagger < 1.$$

Proof. Because C is real symmetric, the spectral theorem gives $C = U \text{diag}(\lambda_j) U^T$ with U orthogonal. The hypothesis $C \prec I$ implies $1 - \lambda_j > 0$, hence

$$(I - C)^{-1} = U \text{diag}((1 - \lambda_j)^{-1}) U^T.$$

Taking the quadratic form with u proves (3). For the second assertion take the Schur complement of the positive leading block $I - C$ in (1):

$$\mathcal{A}/(I - C) = 1 - u^T (I - C)^{-1} u = 1 - q_{\text{full}}^\dagger.$$

The Schur-complement criterion proves the equivalence. $\square$

Theorem 2 (Two-band sufficient criterion). *Under the assumptions of Theorem 1, fix $\theta < 1$ and let $P_\theta = \mathbf{1}_{(\theta, 1)}(C)$. Writing $\lambda_{\max} = \lambda_{\max}(C)$, one has*

$$q_{\text{full}}^\dagger \leq \frac{\|(I - P_\theta)u\|^2}{1 - \theta} + \frac{\|P_\theta u\|^2}{1 - \lambda_{\max}}. \quad (4)$$

Consequently, strict inequality of the right-hand side with 1 is a sufficient certificate for the race inequality.

Proof. Split the sum (3) into eigenvalues $\lambda_j \leq \theta$ and $\lambda_j > \theta$. On the first set, $(1 - \lambda_j)^{-1} \leq (1 - \theta)^{-1}$. On the second, $(1 - \lambda_j)^{-1} \leq (1 - \lambda_{\max})^{-1}$. Parseval's identity turns the two sums of squared coefficients into $\|(I - P_\theta)u\|^2$ and $\|P_\theta u\|^2$. This proves (4); combine it with Theorem 1 and (2). $\square$

Proposition 1 (The fixed-mesh blocks are finite). *For every integer $r \geq 0$,*

$$\{k \in \mathbb{N} : \lfloor \sqrt{k} \rfloor = r\} = \{r^2, r^2 + 1, \dots, (r + 1)^2 - 1\}$$

and this set has cardinality $2r + 1$. On it the selected spacing is

$$\Delta_k = \frac{\log a_k}{m_k + 1} = 2^{-r} \log 2.$$

Thus a monotonicity or averaging argument confined to one fixed r cannot by itself establish an infinitely-often statement.

Proof. The equality $\lfloor \sqrt{k} \rfloor = r$ is equivalent to $r \leq \sqrt{k} < r + 1$, hence to $r^2 \leq k < (r + 1)^2$. There are $(r + 1)^2 - r^2 = 2r + 1$ such integers. Since $a_k = 2^k$ and $m_k + 1 = k2^r$ on this block,

$$\Delta_k = \frac{k \log 2}{k2^r} = 2^{-r} \log 2.$$

Every one of these blocks is finite, which proves the quantifier statement. $\square$

3 Numerical adjudication of the three attacks

The companion probe uses the complete prime-power comb through $a_k^2 = 4^k$ and the same fold, border, chain, and solve as r459. The spectral identity agrees with the direct solve to $3.85 \cdot 10^{-14}$ and with the r459 race object to $1.89 \cdot 10^{-14}$.

k	N	race	$\lambda_{\max}(C)$	raw bound	best two-band	Gersh.	top overlap
5	10	.675060	.8269525	4.825	.910	-1.284	$9.53 \cdot 10^{-3}$
6	12	.694907	.9446612	14.227	1.319	-1.404	$2.22 \cdot 10^{-6}$
7	14	.628736	.9867839	47.207	1.167	-1.260	$8.89 \cdot 10^{-4}$
8	16	.606566	.9941763	93.728	1.726	-1.162	$5.05 \cdot 10^{-4}$
9	36	.635369	.9991328	724.586	4.382	-2.274	$1.94 \cdot 10^{-5}$
10	40	.757783	.9995864	1593.258	5.948	-2.634	$4.15 \cdot 10^{-6}$
11	44	.715782	.9998267	2755.648	6.651	-2.122	$5.10 \cdot 10^{-6}$
12	48	.693564	.9999128	4991.649	10.831	-1.975	$1.54 \cdot 10^{-6}$

Attack 1: norm of the perturbation. The canonical one-number resolvent estimate is

$$q_{\text{full}}^{\dagger} \leq \frac{\|u\|^2}{1 - \lambda_{\max}(C)}. \quad (5)$$

It already gives $4.825 > 1$ at $k = 5$ and worsens to 4991.649 at $k = 12$. The reason is visible rather than conjectural: $\lambda_{\max}(C) \uparrow 1$, while the squared overlap of u with the top eigenvector is only $1.5 \cdot 10^{-6}$ – $9.5 \cdot 10^{-3}$; the top mode contributes at most 5.6% of the true q on this census. The operator norm discards precisely this alignment. Even optimizing Theorem 2 over every spectral cut certifies only $k = 5$ ($0.910358 < 1$); its best value is > 1 on $k = 6, \dots, 12$.

A second obstruction prevents the proposed Weyl comparison with the ARCH baseline: q_* is computed at the smaller degree J_P , whereas $q_{\text{full}}^{\dagger}$ is at degree N . Weyl requires two operators on the same space. The full-depth ARCH signed chain is not a positive baseline; it has respectively 3, 3, 4, 6, 6, 7, 8 sign flips on $k = 5, \dots, 12$. Thus the missing extension from J_P to N would itself contain the positivity problem one hoped to prove.

Attack 2: fixed r . The ARCH prefix values are numerically identical within the two observed blocks:

$$q_* = 0.624231952161870 \quad (k = 5, \dots, 8), \quad q_* = 0.719102723128236 \quad (k = 9, \dots, 12).$$

This useful simplification does not address the quantifier: Proposition 1 proves that every such block is finite. An infinite-subsequence proof must compare infinitely many different r and hence infinitely many different spacings.

Attack 3: direct positivity. For every tested main window the true augmented minimum eigenvalue is positive and $\rho_{N-1} < 5/7$, consistently with r459. But the Gershgorin lower bound of (1) lies between -2.634 and -1.162 on all eight rows. Interlacing alone gives no sign because the parent ARCH cap is indefinite. Direct positivity therefore returns to the spectral alignment in Theorem 1; no source-pure Markov/interlacing inequality was obtained.

4 Exact remaining gap

Remaining gap (verbatim). Prove, from the explicit von-Mangoldt tents and smooth-border source and without evaluating q_k itself, that for infinitely many selected k one has $C_k \prec I$ and there exists $\theta_k < 1$ such that

$$\sum_{\lambda_{k,j} > \theta_k} \frac{|\langle u_k, e_{k,j} \rangle|^2}{1 - \lambda_{k,j}} < 1 - \sum_{\lambda_{k,j} \leq \theta_k} \frac{|\langle u_k, e_{k,j} \rangle|^2}{1 - \lambda_{k,j}}.$$

Equivalently: bound the spectral measure of the explicit normalized border vector near the saturating edge $\lambda = 1$ strongly enough that its Stieltjes transform at 1 is < 1 infinitely often. No such source-derived overlap estimate is proved here.

This statement is deliberately more precise than “control the primes”: it identifies the required currency and the quantifier. It is also honest about its relation to the target: by Theorem 1 it is exactly the missing inequality, split into safe and dangerous spectral bands. The split is useful only if its two terms can be bounded before the target is read back.

5 Alias and refutation audit

Route	classification	reason
Raw norm (5), Gershgorin	acknowledged alias / refuted	Same unsigned-majorant failure class as r387/r429; the probe records its failure and does not advertise it as a mechanism.
Rosser–Schoenfeld or other first-kind ψ upper bounds	not used to claim progress	Inserted only through unsigned mass or operator norm, such bounds erase the overlaps in (3); they cannot repair the measured failure of (5) without additional signed structure.
Fixed- r plateau	new quantifier audit, not octave additivity	No increments are added; the exact cardinality $2r + 1$ shows why one plateau cannot provide the required infinite subsequence.
Spectral projector identity	rigorous reduction, not r429 triangle	It retains signed eigenvector alignment. Without a source estimate for the projector mass it remains a reduction, not a proof.
Potapov, de Branges, Abel, PNT-density replacement	not repeated	No product index, phase dominance, left–right split, or replacement density enters the argument.

Claim boundary

This note proves finite linear-algebra statements and reports a finite full-comb census. It does not prove the race lemma for an infinite subsequence. Nothing here is evidence for or against the Riemann Hypothesis in either direction. No RH claim. No anti-RH claim.